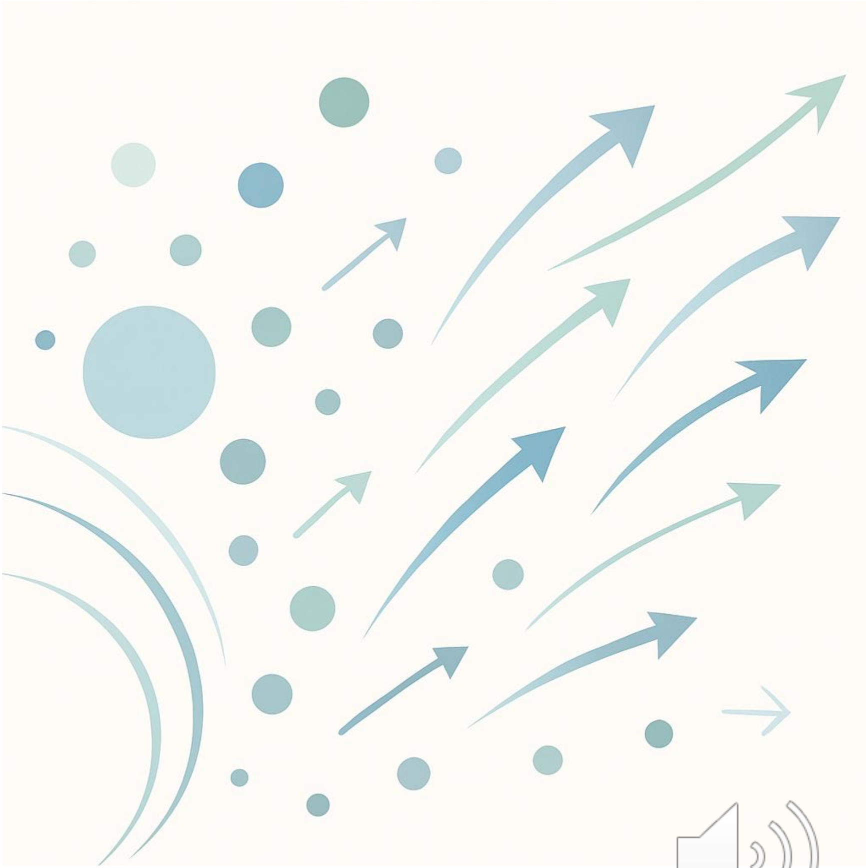
Predicting Employee Attrition

IBM HR Analytics Employee Attrition & Performance

August 7, 2025



Presentation Overview

Introduction

Data Overview

Methods

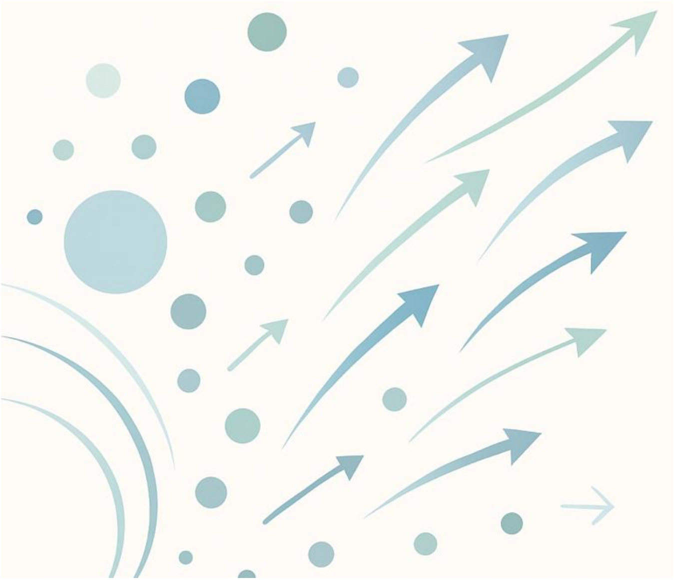
Results

Ethical Considerations

Conclusion

Introduction

Employee attrition disrupts teams, increases recruitment costs and diminishes institutional knowledge.
Predictive analytics can identify at–risk employees and enable HR interventions.



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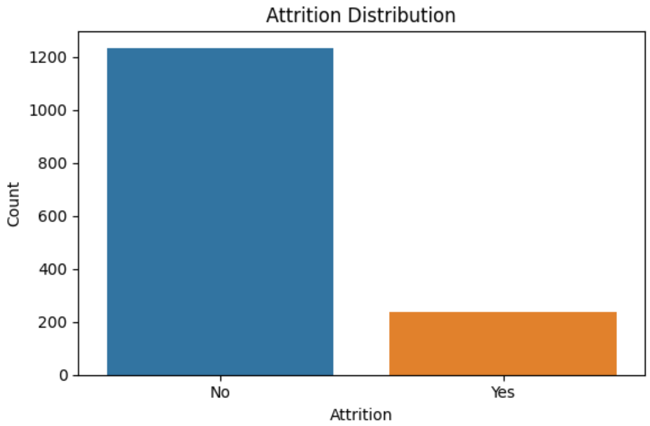
Dataset Overview

1,470 employees
35 original features; about 50 numeric predictors after one-hot encoding
Demographic, job role, satisfaction and compensation variables
Attrition rate ≈16%



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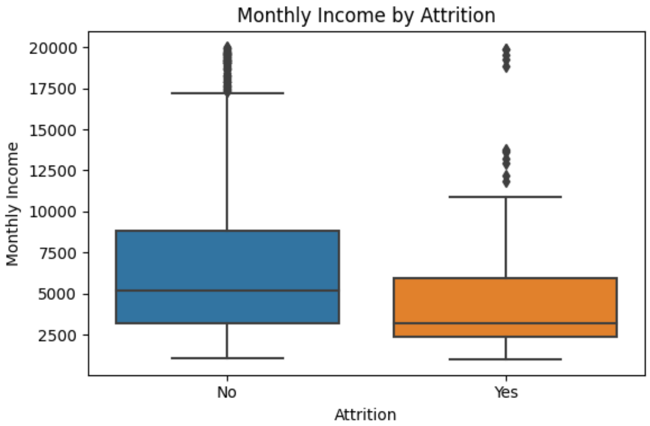
Exploratory Analysis

Employees who left tended to have lower monthly income. Boxplots reveal a right–skewed distribution for employees who stayed, indicating greater earning potential.



Methods

Preprocessing & Resampling

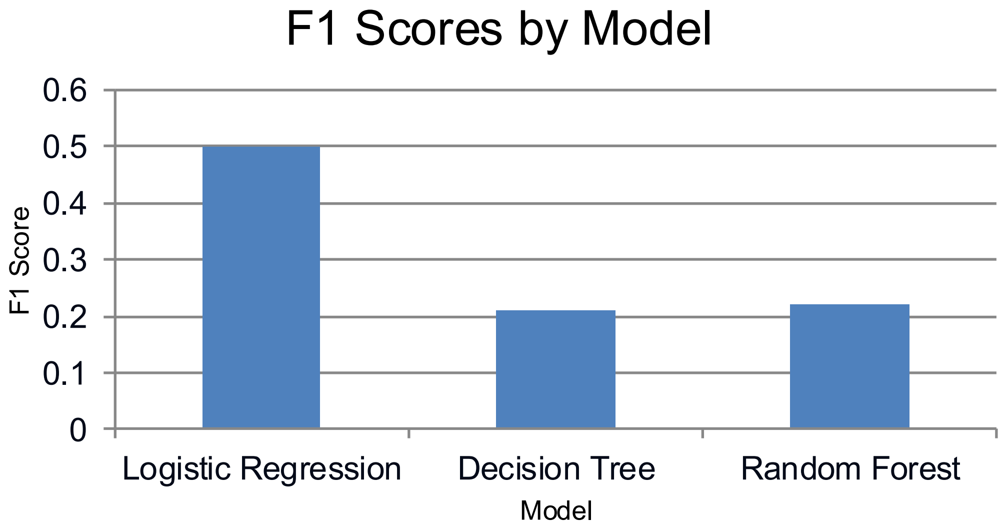
- Drop identifiers and constant columns
- One-hot encode categorical variables
- Stratified 80/20 train–test split
- SMOTE oversampling to balance the minority class

Models

- Logistic Regression: linear classifier with sigmoid activation
- Decision Tree: interpretable rule-based splits
- Random Forest: ensemble of randomised trees

[2] [3]

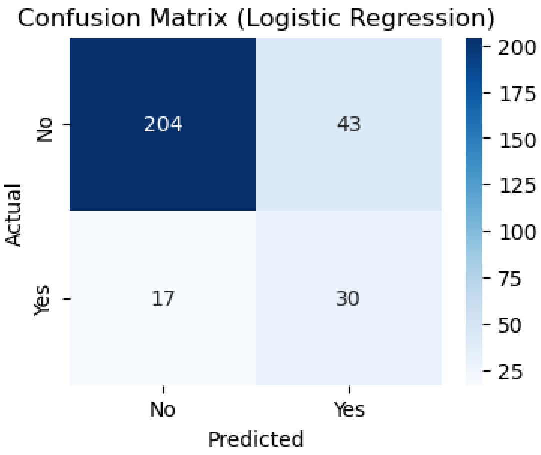
Model Performance



- Logistic regression achieves the best balance of precision and recall.
- Random forest attains high accuracy but suffers from very low recall.
- Decision tree underperforms and exhibits poor sensitivity.

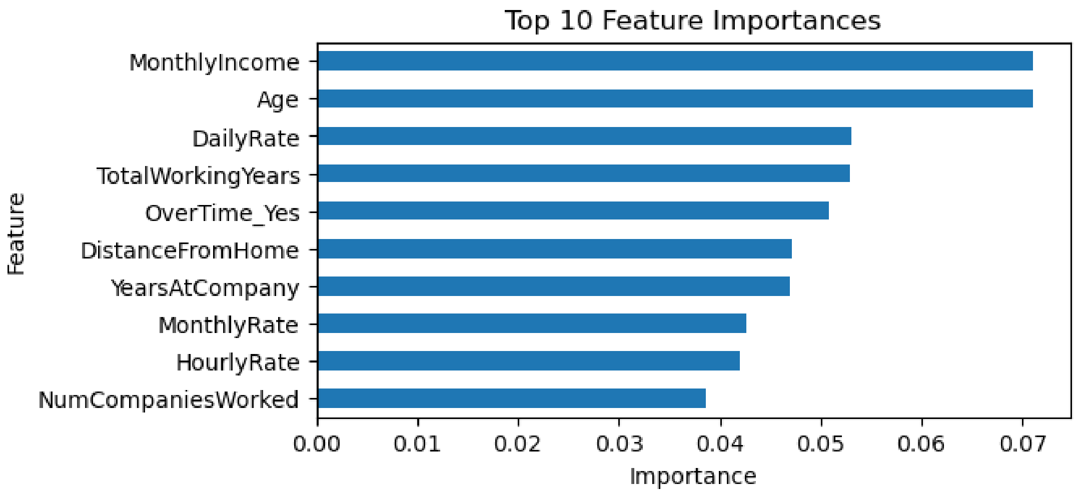
[2] [3]

Confusion Matrix



Confusion matrix for logistic regression (test set)

Important Predictors



- Age and monthly income emerge as strong predictors of attrition.
- Longer tenure and higher pay reduce the likelihood of leaving.
- Working overtime still raises attrition risk.

[3]

Ethical Considerations

- Assess fairness across demographic groups
- Protect privacy and secure data storage
- Communicate openly about model use

[6]

Conclusion & Next Steps

- Logistic regression offers the best trade-off between precision and recall.
- Age and monthly income are among the strongest predictors, while overtime still elevates risk.
- Random forest attains high overall accuracy but fails to identify many leavers; decision trees underperform.
- Future work: test advanced resampling like SMOTE, incorporate behavioural metrics and run cross-validation.
- Use models to inform—not replace—HR judgement.

[4] [6]

References

[1] Employee attrition reduces profit margins and predictive models can identify at-risk staff. (inseaddataanalytics.github.io)

[2] Logistic regression uses a sigmoid function to predict binary outcomes. (www.geeksforgeeks.org)

[3] Random forests are ensembles of decision trees grown on random subsets. (www.geeksforgeeks.org)

[4] SMOTE creates synthetic minority samples by interpolating between neighbours. (pmc.ncbi.nlm.nih.gov)

[5] Predictive HR analytics should avoid bias, protect privacy and be transparent. (www.hrcloud.com)